SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Hackstreet Boys
Project / Solution Name	Stock Manager
Name (Leader)	Josiah Chong Ren Shun
Email & Phone (Leader)	josiahc159@gmail.com (+60 16-843 4553)
University Affiliation(s)	Xiamen University Malaysia, Monash University Malaysia
Live Prototype URL	**https://sme-inventory-workflow-backend.replit.app/team-preview/**
Code Repository URL (public / access granted to judges)	**https://github.com/Adilkrab/Hackstreet-Boys**

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Stock Manager is a three-part web application hosted on Replit.

1. User interface: a React frontend (built with Vite) with five screens: dashboard, products, stock count, approvals and activity log. A role switch simulates login for the Owner and the Stock clerk.
2. Backend: a Python FastAPI REST API validates every request and runs the approval rules. A count differing by 2 units or fewer and under RM50 is corrected automatically; anything larger, negative or outdated waits for the owner.
3. Database: SQLite, accessed through SQLAlchemy, stores products, stock movements, adjustment requests and the audit log. Each stock change and its audit entry are saved in a single transaction.
4. Integration: a simulated Orbyte sync endpoint packages approved changes and low-stock alerts as JSON and logs them. It makes no external calls.

The frontend calls the backend over HTTP (/api routes). No AI model or external API is called at runtime.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
`FastAPI ≥0.115`	Backend REST API framework	MIT

Name & Version	Purpose in Project	License
`Uvicorn ≥0.34`	Web server that runs the backend	BSD-3-Clause
SQLAlchemy 2.x	Database models and transactions	MIT
SQLite 3	Embedded database	Public domain
Pydantic 2.x	Request and response validation	MIT
`python-multipart ≥0.0.20`	CSV / XLSX file uploads	Apache-2.0
`openpyxl ≥3.1.5`	Reading Excel (.xlsx) imports	MIT
`pytest ≥8`	Automated backend tests	MIT
`httpx ≥0.28`	Test client for API tests	BSD-3-Clause

Name & Version	Purpose in Project	License
React 19 / React DOM 19	Frontend user interface	MIT
Vite 6	Frontend development server and build	MIT
@vitejs/plugin-react 4	React support for Vite	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None	No external APIs are called at runtime.	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts and runs the live prototype (frontend and backend); public demo link	Replit Core (sponsor-provided) / managed by Replit; region not displayed
GitHub	Source code repository and version history	Free / not applicable

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	Not applicable	No AI model is embedded or called at runtime.	Not applicable

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo minimarket products and demo stock scenarios **Path:** src/sample_data/products.csv	Team-created, in our GitHub repository	Synthetic	Apache-2.0 (project license)	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude (Anthropic) Opus via claude.ai (exact version not recorded)	React frontend support, code-review fixes, and drafting the proposal and technical declaration.	Moderate
Codex / ChatGPT (OpenAI) GPT-6 via Codex	Backend and integration review, submission structure and documents, README updates, and pitch-deck preparation.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviewed AI-assisted code and text before committing it. Generated code was checked against the documented API and business rules, run with the real backend, and tested end to end as both Owner and Stock clerk. Automated backend tests and the production frontend build were run after integration. Code-review findings covering import preview, CSV export, pagination, error handling, stale approvals and rollback were corrected and retested. The team accepts responsibility for the final code, documentation and submitted media.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core plan (platform and credits)	Hosts the live prototype: runs the FastAPI backend and React frontend and provides the public link for judges.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replaceable. Replit hosts the live prototype and gives judges a public link, but the app is a standard FastAPI and React project with no Replit-specific code. It runs the same way on a laptop or any other host that can run Python and Node.js. RunPod credits were not used, because the prototype has no GPU workload.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No real user data is collected. The prototype runs only on synthetic, team-created data: made-up products, a fictional shop and fictional staff names. The only personal-looking fields are the names typed when someone records a count, approval or stock change; these are attribution labels in the activity log, not verified identities.

Data is stored in a SQLite file on the Replit project's disk and is not sent to any third-party service or AI API. The demo data can be wiped and re-created at any time with the seed script. API keys and passwords are not needed and none are stored in the repository.

In a real deployment we would follow Malaysia's PDPA principles: collect only staff names and roles needed for accountability, obtain consent, use real authentication, restrict audit-log access to the owner, encrypt data in transit and set a retention period.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The prototype has no runtime AI, so hallucination and model-bias risks do not currently apply. Main limitations:
1. Role switching is simulated and names are unverified. The UI hides approval controls from Clerk view, but production authentication and server-side authorization remain future work.
2. Orbyte sync generates and logs a payload without sending it externally.
3. Auto-fix thresholds (2 units, RM50) are fixed; future versions should allow per-shop settings.
4. SQLite suits a single-shop demo. Transactions and stale-count checks reduce consistency risk, but production needs durable managed storage.
5. Planned pattern detection and summaries will remain advisory; the owner keeps the final decision.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.

- [x] Any use of official SDC sponsor resources has been declared in Section 5.

- [x] The prototype described here is accessible to judges through the links provided in Section 1.

- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

October 24th 2026